

Sravani Peddi

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SUMMARY

Data driven decision maker with expertise in Python, SQL, MongoDB, and Data Structures. Skilled in Data Visualization using Power BI and Tableau, with experience in machine learning algorithms like clustering, regression and classification. Proficient in leveraging analytical techniques to derive insights and drive data-driven decisions.

EDUCATION

Master Of Science in Computer Science May 2025
Wright State University, Dayton, OH GPA:3.6

Bachelor Of Technology in Information Technology May 2023
Vasireddy Venkatadri Institute of Technology GPA:3.85

WORK EXPERIENCE

Junior Software Engineer Intern, LexTech Inc May 2024 – August 2024

- Engineered a Python data pipeline extracting 500,000+ crack growth data points from AFGROW output files, enabling batch processing and reducing manual analysis time by 40%
- Developed statistical models using Pandas/NumPy to calculate stress-intensity factors (ΔK) and predict component lifespan via Paris' Law coefficients
- Implemented anomaly detection algorithms that identified 15% more irregular crack patterns in alloy test data compared to manual review
- Automated legacy Excel workflows with VBA, reducing reporting time by 30% while migrating processes to Python
- Created interactive Matplotlib/Seaborn dashboards visualizing:
 - Crack propagation trends (a-N curves)
 - Critical stress thresholds for aircraft components
 - Environmental degradation effects on material fatigue
- Conducted fracture mechanics analysis that improved lifespan prediction accuracy by 20% for high-risk structural parts.

CERTIFICATIONS

- Hacker Rank – SQL Gold Badge** May 2025
- Quantium Data Analytics Simulation** December 2024
- Microsoft Azure AI Fundamentals Certificate** March 2023
- AWS Cloud Virtual Internship by AICTE** May 2022

TECHNICAL PROJECTS

Wine Dataset Correlation Analyzer March 2025

- Built an interactive GUI tool that uncovers hidden relationships in wine characteristics—from acidity to alcohol content—helping enthusiasts and sommeliers make data-driven tasting decisions.
- Engineered dynamic visualizations (Matplotlib/Seaborn) to reveal surprising correlations, like how tannin levels might predict wine body, with bulletproof error handling for messy real-world data.
- Went beyond basic Pandas. Corr() by integrating live MySQL queries, allowing users to slice data by region, vintage, or grape type on the fly.

Weather Data Analysis

December 2024

- Transformed raw meteorological data into actionable insights, detecting microclimate anomalies (e.g., "Why does downtown heat up 3°F faster than suburbs?") using custom Python anomaly detection scripts.
- Designed a Tableau dashboard that tells a story—not just charts—blending historical trends with real-time alerts, used by local farmers to optimize crop rotations.
- Uncovered counterintuitive patterns, like rainfall inversely affecting solar efficiency in coastal areas, through multivariate time-series analysis.

SKILLS

Data Acquisition & Preparation: data structures, data warehousing, ETL processes, data extraction, cleansing, parsing, and transformation.

Data Visualization Tools: Power BI, Tableau and python (Matplotlib, Seaborn)

Programming Languages: Python, VBA, SQL

Databases: MongoDB, MySQL

Cloud & DevOps: AWS services: EC2, S3, IAM roles, CI/CD pipeline management using Jenkins

Tools & Automation: Microsoft Excel: Pivot tables, VLOOKUP, VBA, Macros, ETL tools and distributed version control systems (GIT, GitHub).